

CS787: Generative Artificial Intelligence

Assignment 1: Total Marks = 50

Due: 12 September, 2026, 11:30 pm

Consider the word embeddings of various Indian languages at:

https://drive.google.com/drive/folders/1PNPdz_mlpHw-wgZsBruSwfYxLZ5edFg7.

Choose your mother tongue for this assignment. (In case embeddings from your mother tongue is absent, choose another Indian language.)

Submit the assignment at hello.iitk.ac.in. If you have written Python code, submit the Notebook file with appropriate comments, analyses and answers. If you have used some other programming language, submit your code with appropriate comments and a Readme.txt file with details on how to run your code along with analyses and answers.

Q1: [15 marks] Think of 10 triplets of words such that two of them are similar and the third one is dissimilar. (An example in English is {'house', 'home', 'zebra'}.) For each triplet, compute the three pairs of distances, using all the 6 embedding types. What do you conclude?

Q2: [10 marks] Repeat the above exercise for English using the word embeddings available from:

<https://fasttext.cc/docs/en/crawl-vectors.html>

and

<https://wikipedia2vec.github.io/wikipedia2vec/pretrained/>.

What do you conclude?

Q3: [15 marks] Think of 10 quadruplets for words such as “king - man + woman = queen”. For the first three words, find the fourth word whose vector is closest to the aim. What do you conclude?

Q4: [10 marks] Repeat the above exercise for English. What do you conclude?